

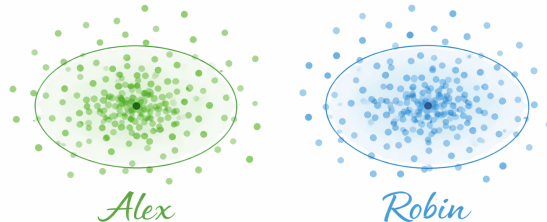
INTRODUCTION TO MACHINE LEARNING

PROJECT WEEK 7

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University of Luxembourg - Summer Semester
2025-2026

April 14, 2026



THE GOAL AND SOME DOMAIN KNOWLEDGE

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- ▶ Genes: How well you break down cholesterole and fat (metabolism)
- ▶ ...

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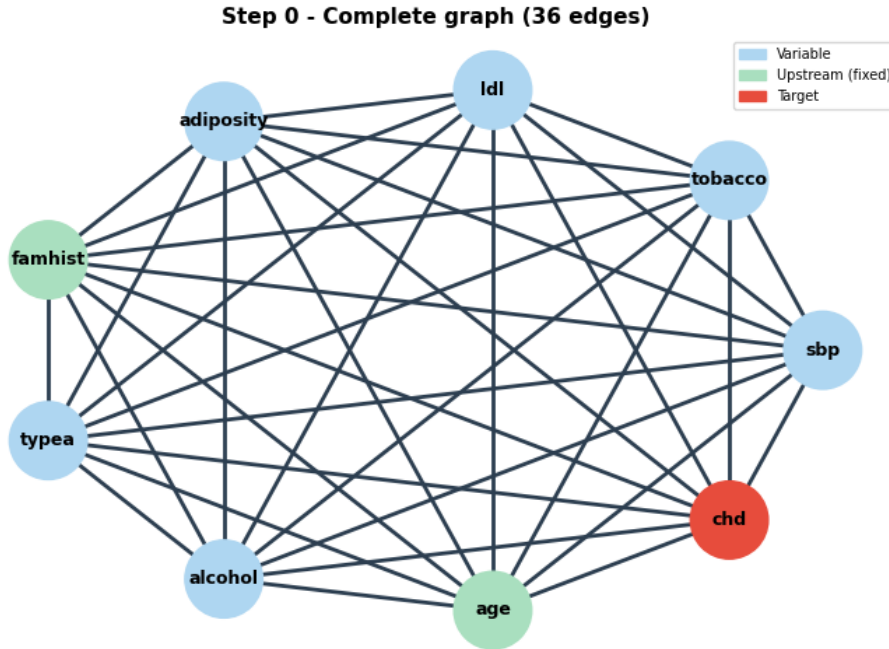
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 - => Last two quite similar: Keep only one

STEP 0: START FULLY CONNECTED

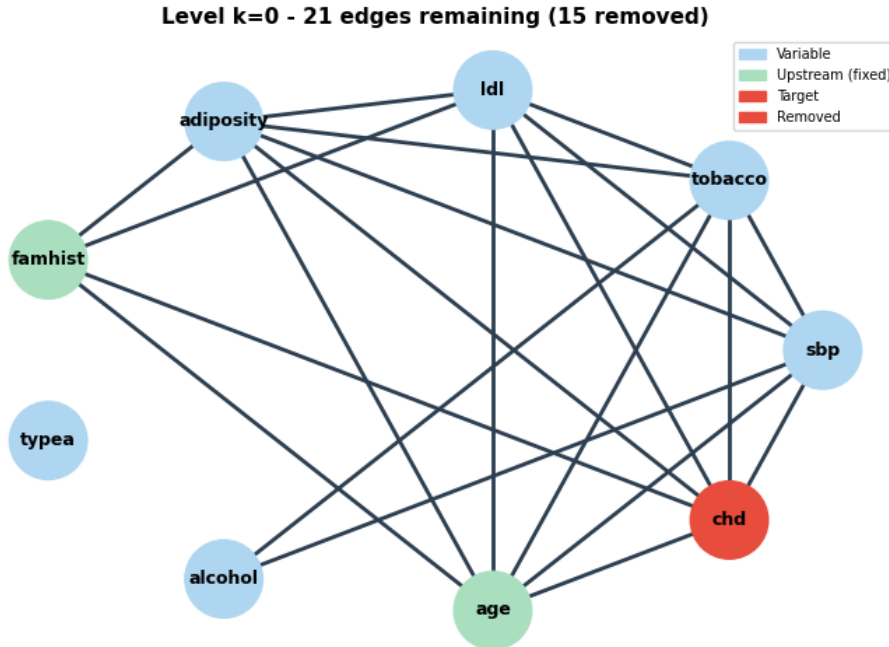
EVERY POSSIBLE EDGE PRESENT



- ▶ **sbp:**
blood pressure
- ▶ **ldl:**
cholesterol
- ▶ **chd:**
heart disease
- ▶ **famhist:**
family history

STEP 1: $K=0$ - UNCONDITIONAL INDEPENDENCE

15 EDGES REMOVED: PAIRS INDEPENDENT WITHOUT CONDITIONING

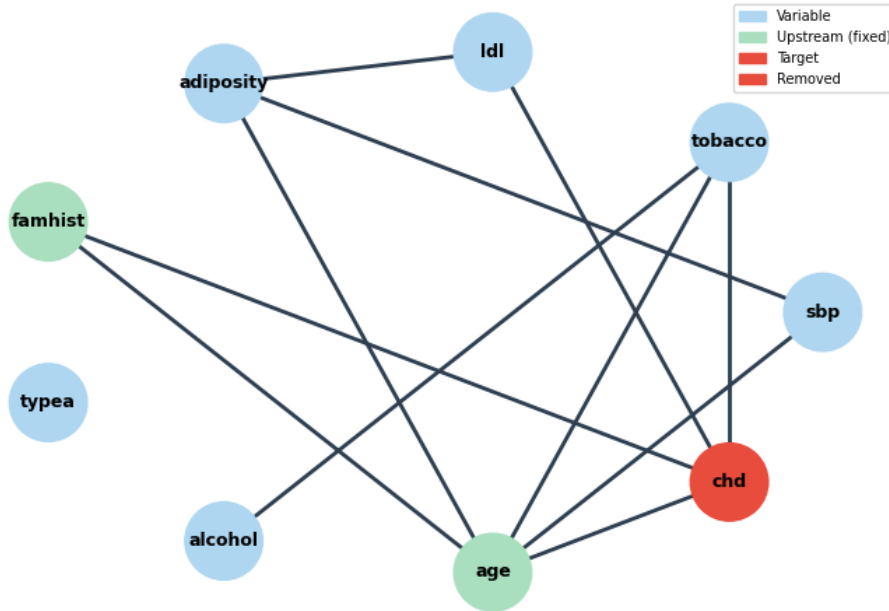


- **sbp:**
blood pressure
- **ldl:**
cholesterol
- **chd:**
heart disease
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STEP 2: $K=1$ - CONDITION ON ONE VARIABLE

10 EDGES REMOVED: PAIRS BECOMING INDEPENDENT ONCE ONE NEIGHBOR IS HELD FIXED

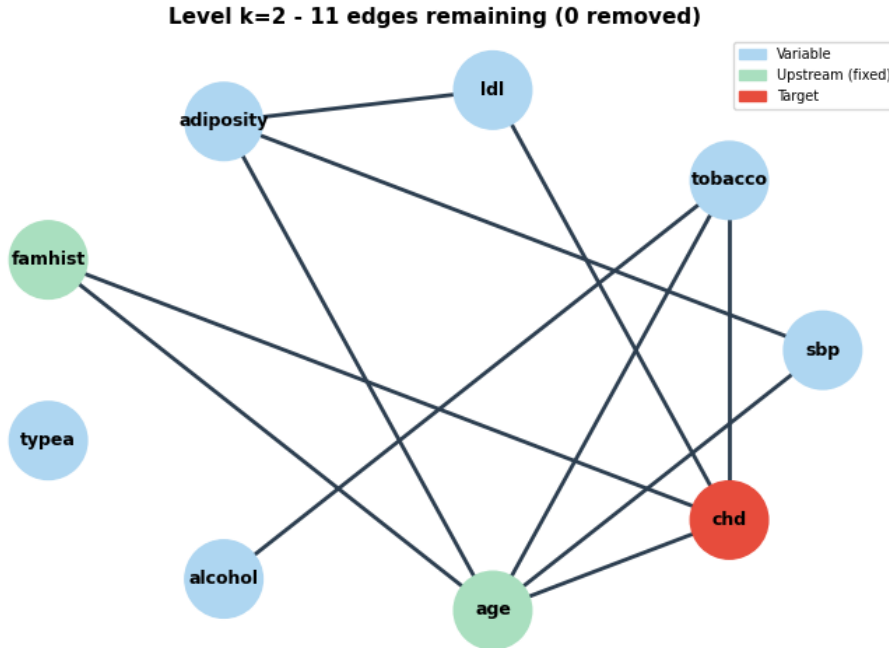
Level $k=1$ - 11 edges remaining (10 removed)



- **sbp**: blood pressure
- **ldl**: cholesterol
- **chd**: heart disease
- **famhist**: family history

STEP 3: K=2 - CONDITION ON TWO VARIABLES

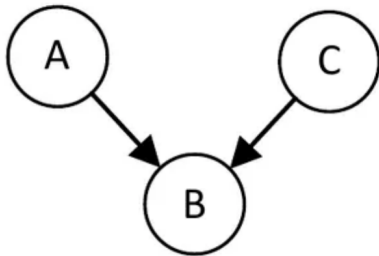
NO EDGE REMOVED: WE STICK WITH K=1



- **sbp:**
blood pressure
- **ldl:**
cholesterol
- **chd:**
heart disease
- **famhist:**
family history

V-STRUCTURES: ORIENT COLLIDERS

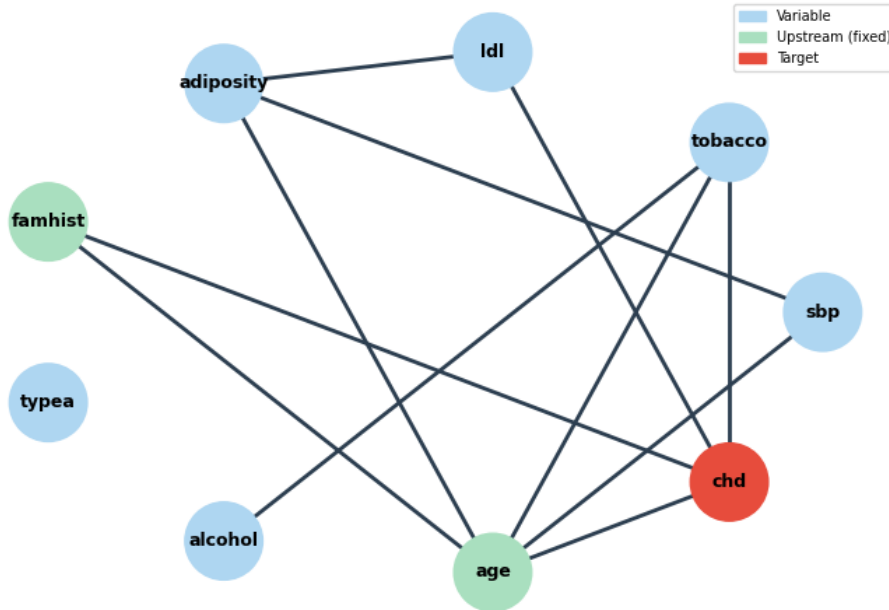
- ▶ Triple $A - B - C$ where A and C are not adjacent, AND
- ▶ A and C became independent without conditioning on B
- ▶ Then B is a collider: orient $A \rightarrow B \leftarrow C$
- ▶ 10 v-structures found in our skeleton



V-STRUCTURE EXAMPLE

ALCOHOL $\rightarrow$ TOBACCO $\leftarrow$ AGE

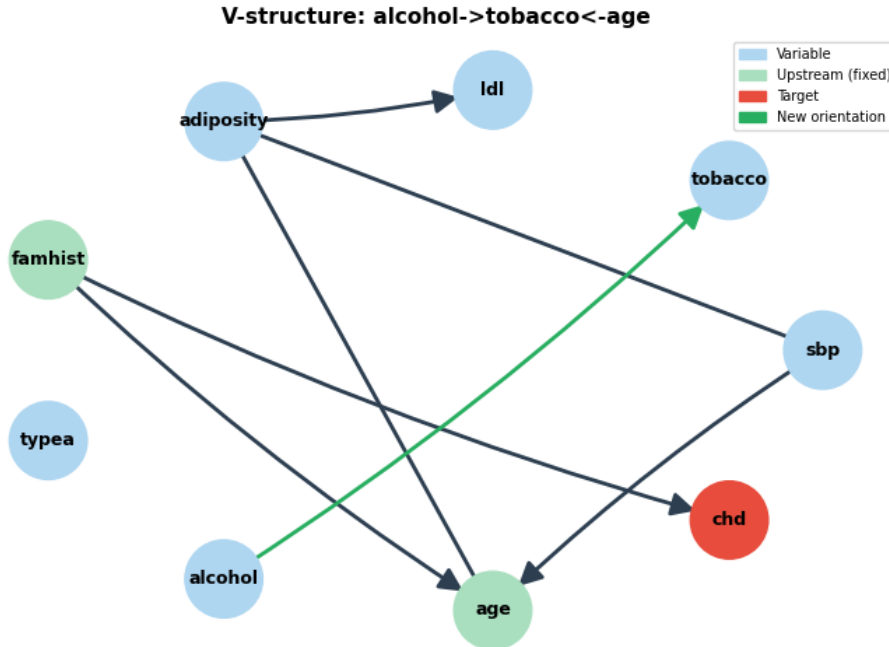
Level k=2 - 11 edges remaining (0 removed)



- Consider node "alcohol"
- Alcohol and age not adjacent
- Yet both predict tobacco
- Tobacco = collider

V-STRUCTURE EXAMPLE

ALCOHOL $\rightarrow$ TOBACCO $\leftarrow$ AGE

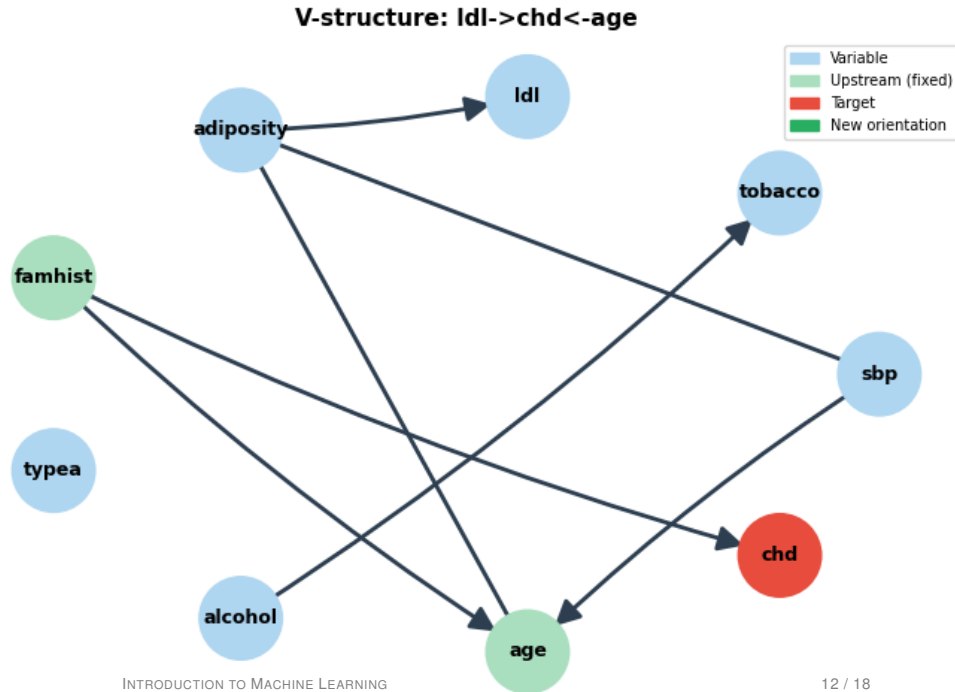


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PAUL-CLARK ALGORITHM

FINAL GRAPH: PC + EXPERT KNOWLEDGE

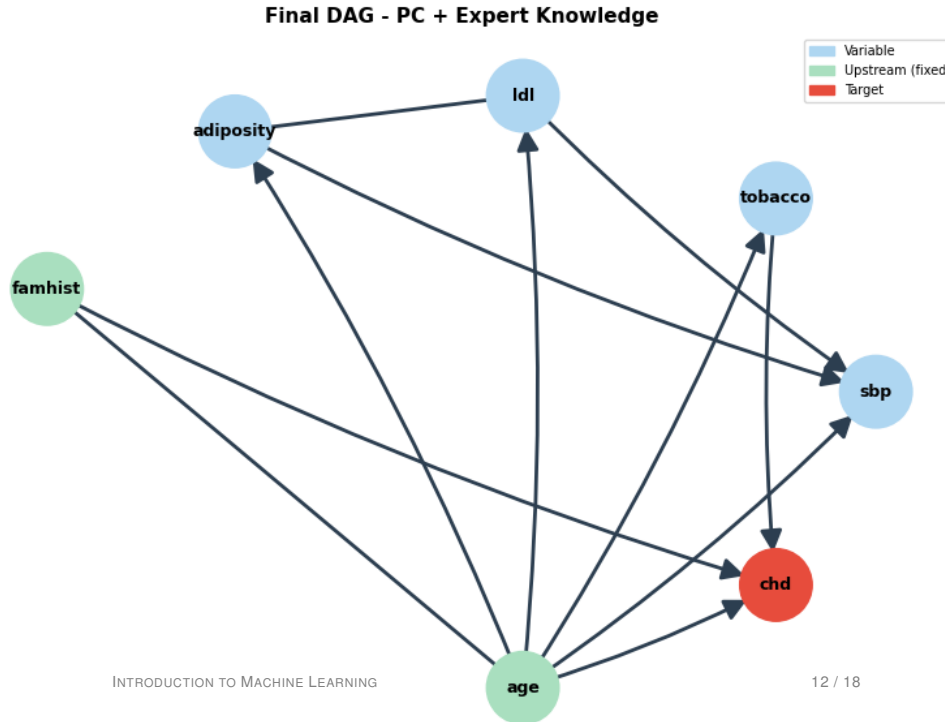
- Some edges not logical:
blood pressure → *age*



PAUL-CLARK ALGORITHM

FINAL GRAPH: PC + EXPERT KNOWLEDGE

- Some edges not logical:
blood pressure → *age*
- Use expert knowledge:
 - Upstream:
age, family history
 - Downstream:
heart disease (chd)



CORRECT DAG EDGES – EVIDENCE

Edge	Source
adiposity → sbp	Hall et al. (2015): Obesity increases blood pressure via vascular and renal mechanisms (Circulation Research)
age → sbp	Franklin et al. (1997): Hemodynamic patterns of age-related changes in blood pressure (Circulation)
age → chd	American Heart Association (2023): Cardiovascular risk increases strongly with age (AHA Statistics Report)
tobacco → chd	U.S. Surgeon General (2014): Smoking is a causal factor for coronary heart disease
famhist → chd	Lloyd-Jones et al. (2004): Family history significantly increases CHD risk (Framingham Heart Study)
adiposity → ldl	Feingold (2021): Obesity alters lipid metabolism and raises LDL levels (Endotext)

MISLEADING EDGES

- ▶ Many edges make sense, align with our "pipe model"
e.g. age, famhist, and tobacco → heart disease

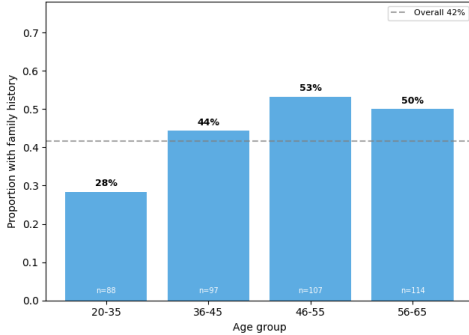
MISLEADING EDGES

- ▶ Many edges make sense, align with our "pipe model"
e.g. age, famhist, and tobacco → heart disease
- ▶ Some misleading:
 - age - family history (neither causes the other)
 - age → tobacco (direction misleading)
 - cholesterol - adiposity (expected: →)
 - cholesterol - blood pressure (largely spurious)

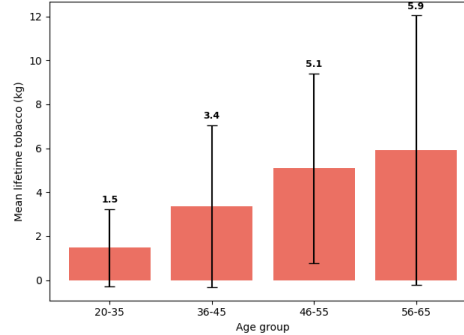
MISLEADING EDGES - DATA PATTERNS

Why the PC Algorithm Gets Confused - Data Patterns in SAheart

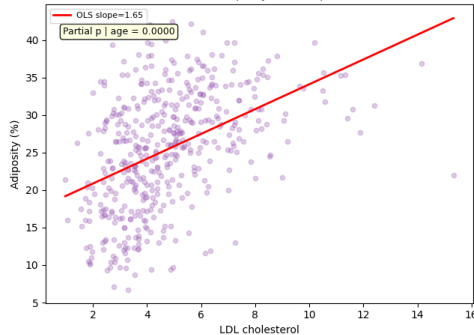
Case 1: age vs famhist $r=0.240$, $p=0.000$



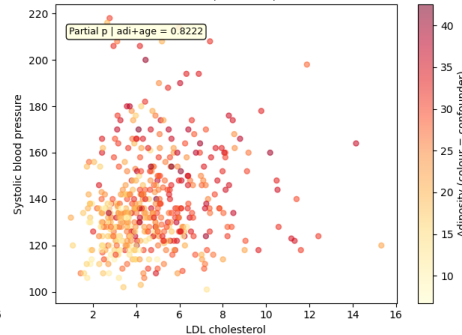
Case 2: age vs tobacco $r=0.450$, $p=1.88e-24$



Case 3: ldl vs adiposity $r=0.440$, $p=0.0000$



Case 4: ldl vs sbp $r=0.158$, $p=0.0006$



1. famhist rate rises with age (ascertainment)
2. tobacco $\approx 4x$ across age groups (cumulative measure)
3. ldl-adiposity symmetric scatter (no v-structure)
4. ldl-sbp colored by adiposity (shared confounder)

FURTHER IMPROVEMENT IDEAS

In expert knowledge:

- ▶ Forbid age - famhist

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- ▶ Forbid age → tobacco
- ▶ Only allow adiposity → cholesterol

FURTHER IMPROVEMENT IDEAS

In expert knowledge:

- ▶ Forbid age - famhist
- ▶ Forbid age $\rightarrow$ tobacco
- ▶ Only allow adiposity $\rightarrow$ cholesterol
- ▶ Cholesterol vs blood pressure:
 - True causal graph: $sbp \leftarrow adiposity \rightarrow ldl$
 - Increase amount of neighbors to condition on (>2)

REFERENCES

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THE END

Thank you for your attention!

Questions?